

Escaping the Valley of Boredom: Efficient User Resizing in Virtual Reality

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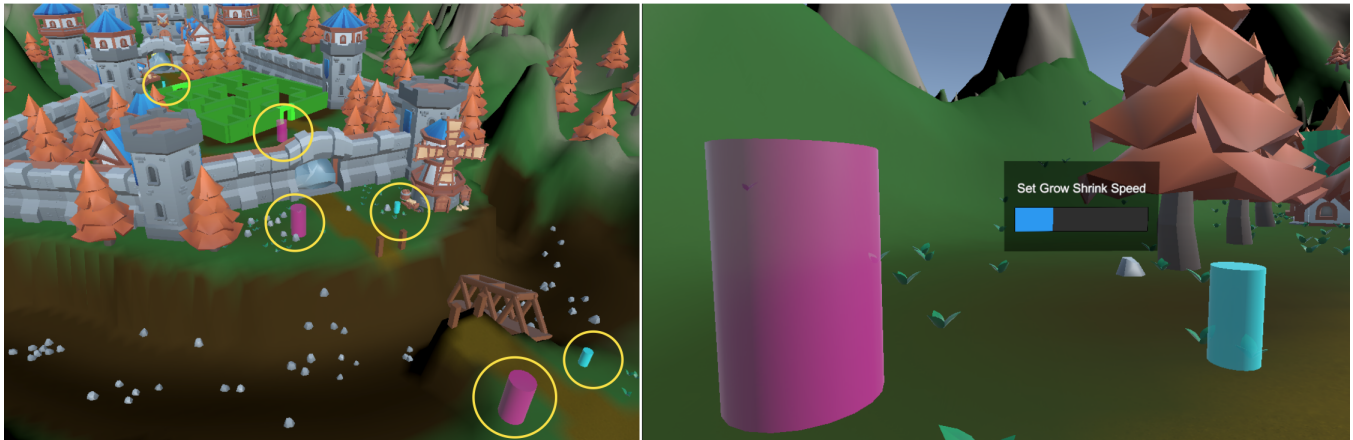


Figure 1: *Left:* A traversal task that elicits repeated grow-shrink events, where participants resize to overcome obstacles (e.g., cross the river, study the maze) and use the small scale to fit below low ceiling obstacles (e.g., maze entrance or castle gate). The user changes their size when entering pink *shrinking* or cyan *growing* cylinders. *Right:* The selection dialog, where participants adjust the grow-shrink transition duration via a 15-step discrete slider to identify their most comfortable setting.

Abstract

Resizing the user’s virtual representation in virtual reality (grow-shrink) can support rapid traversal, overview, and size-dependent interaction. However, the transition between scales introduces risks of sensory conflict and motion sickness. We conducted a within-subjects user study (N=15) to evaluate the effects of transition duration and preparedness on user comfort and preference. Results indicate that in our sample neither option significantly increases motion sickness, suggesting that this technique is a viable alternative to common locomotion methods. However, we identified a strong “Efficiency Bias”: users consistently preferred sub-second transitions over “cinematic” slow transitions (10s). Our findings suggest a *Valley of Boredom* for intermediate transition durations (2–5s), leading to design guidelines that prioritize either instantaneous “Snap” scaling for efficiency or slow “Glide” scaling for accessibility, while avoiding the uncomfortable middle spectrum.

CCS Concepts

• **Human-centered computing** → *Empirical studies in HCI*; **Virtual reality**; *Laboratory experiments*.

Keywords

Virtual Reality, Locomotion, Cybersickness, Multiscale Navigation, Player Resizing

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1 Introduction

Virtual reality (VR) enables experiences and interaction styles that are difficult or impossible in the physical world. One such interaction is dynamically resizing the user’s virtual representation. By shrinking or growing, users can switch between precise interaction and a global overview. Dynamic scale change is helpful e.g., in VR games, multiscale data visualization, and training or design applications, where users benefit from switching efficiently between detailed interaction and a global overview.



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At the same time, viewpoint and locomotion effects are typical drivers of cybersickness in VR [3]. Resizing transitions are a distinctive case: they alter perceived scale and can introduce sustained changes in visual motion cues during the transition. Designers implementing grow-shrink mechanics must decide how scale changes unfold (e.g., transition duration, motion profile) and what information or agency users have during transitions. Two plausible comfort drivers are (i) the duration of the scale transition and (ii) the user's preparedness for the upcoming change. Moreover, comfort may evolve over repeated exposures, and such adaptation may differ by prior VR experience.

This paper contributes a user study of grow-shrink transitions in VR. We have built a gamified VR environment primarily as a study apparatus to (a) elicit repeated grow-shrink events in structured tasks and (b) manipulate transition duration and preparedness cueing in a controlled way. We address three questions:

- (1) How does the duration of the transition affect cybersickness?
- (2) Does preparedness cueing (with or without warning) reduce cybersickness during resizing?
- (3) Which transition durations do users perceive as most comfortable when given control?

We also explore how prior VR experience moderates these effects over repeated exposures.

2 Related Work

Locomotion techniques in VR often involve trade-offs between presence, efficiency, and comfort. Changing the user's scale induces the **Alice in Wonderland Effect** (AIWE) [2], which has been studied in multiscale navigation and interaction [1, 8, 16] and appears in VR games. The AIWE describes how users perceptually embody differently sized virtual bodies, altering perceived environment scale [10]. Because resizing alters perceived spatial relationships and induces continuous visual transformation of the environment, it functions as a form of artificial locomotion. However, systematic evaluation of grow-shrink transition dynamics—particularly how transition duration and cueing influence user comfort—remains limited. This work operationalizes grow-shrink transitions as repeated interaction events and evaluates controllable transition parameters relevant to system design.

Cybersickness [3] arises primarily from sensory conflict between visual, vestibular, and proprioceptive signals, and is commonly assessed using self-report instruments. Sensory Conflict Theory posits that sickness occurs when sensory inputs violate the brain's internal predictions [13]. This relationship is illustrated by the Driver–Passenger effect, in which actively initiated and predictable motion produces less sickness than passive motion due to improved feed-forward prediction [14]. In grow-shrink transitions, visual motion is externally imposed, potentially reducing predictive control and increasing discomfort. Transition duration directly affects the intensity of visual motion signals, while preparedness cues may improve predictability by informing users of upcoming changes.

Prior work has explored mitigation strategies such as limiting acceleration, reducing optical flow, and improving user control and predictability [5, 12, 15]. Anticipatory cues, including simple audiovisual warnings, have been shown to improve predictability

and reduce motion sickness in virtual environments [4, 9]. Motivated by these mechanisms, we evaluate transition duration and preparedness cueing as theoretically grounded design parameters influencing cybersickness and user comfort during grow-shrink transitions.

3 Study Apparatus and Tasks

We developed a VR environment that implements a grow-shrink mechanic, changing the user's scale between two states (regular and large), and supports size-dependent traversal and interaction. The environment is designed to evoke multiple resizing events while participants complete simple tasks.

Resizing interaction. Participants trigger resizing voluntarily by interacting with in-world objects (cylinders) placed at positions that make resizing necessary to progress. During resizing, user movement is locked to avoid confounding effects observed in early testing when participants moved while scaling. The resizing animation uses a linear scale change profile to avoid additional acceleration/deceleration phases.

Task environments. The user study consisted of five levels:

- Two **Duration levels**, structurally similar traversal levels that differ in transition duration (10s or 3s). The levels contain obstacles such as heavy objects, a collapsed bridge, or a maze that require growing and shrinking to proceed.
- Two **Preparedness levels** where the user's size changes at random intervals while they perform a repetitive task (e.g., finding/pressing buttons). In the “warning” condition, an audiovisual cue appears shortly before each size change; in the “no warning” condition, size changes occur without cueing.
- One **Choice level** where participants can adjust resizing duration using a discrete 15-step slider from 0.75s (fastest) to 10s (slowest). Participants could explore various durations and pick the one they find most comfortable.

4 Method

We employed a within-subjects design with two factors: transition duration (10s or 3s) and preparedness (with or without warning).

To mitigate ordering and learning effects, we used a fully balanced blocked design. The tasks were tied to each research question, grouped, and the condition order varied across participants.

Measures. We collected both quantitative and qualitative data:

- **VRSQ (primary outcome)** [7] was administered after each condition-level interaction to assess VR sickness symptoms.
- **VIMSSQ (covariate)** [11] captures individual susceptibility to visually induced motion sickness to control baseline differences.
- **Duration preference logs** from the choice level (up to three adjustment points).
- **Interviews** capturing reasons for duration preference and open-ended descriptions of discomfort.

Procedure. Figure 2 shows the experiment structure. Participants completed a tutorial on controls, followed by the duration and preparedness levels (order counterbalanced), and reported VRSQ after each. They then completed the choice level to select a preferred

resizing duration. A few hours after the session, participants completed a second VRSQ to assess potential negative aftereffects [6].

Participants: We recruited 15 participants (ages 20–34; 5 female, 10 male), primarily university students recruited via leaflets and online postings. Participants reported varying levels of motion sickness susceptibility and limited prior VR experience.

5 Analysis

We analyzed condition effects using linear mixed-effects models (LMMs) with random participant intercepts to account for within-subject structure and baseline differences. VIMSSQ scores were included as a covariate to control individual susceptibility. For the transition duration analysis, residual diagnostics motivated log transformation of VRSQ; for preparedness, diagnostics did not indicate major assumption violations. To examine exposure effects, we modeled VRSQ as a function of chronological interaction index (1–5), VR experience, and their interaction, again using random intercepts and log-transformed VRSQ.

6 Results

Transition duration and preparedness effects. Figure 3 (Left) shows substantial individual variability but no consistent within-participant shift in either duration or preparedness conditions.

The fixed effect of transition duration condition on log-transformed VRSQ was not significant ($\beta = 0.008$, $SE = 0.45$, $t(14) = 0.02$, $p = .986$, 95% CI $[-0.88, 0.90]$). Back-transformed, this corresponds to an estimated 0.8% difference between high- and low-speed. A jackknife cross-validation indicated that outliers did not drive this result.

Preparedness cueing did not show a significant effect on VRSQ ($\beta = -0.47$, $SE = 7.02$, $t(14) = -0.07$, $p = .948$). A one-sided test in the direction of higher scores under warning was also not significant. Refitting the model without the susceptibility covariate yielded the same conclusion.

Repeated exposure and experience-dependent adaptation. Participants' qualitative reports suggested that discomfort changed over time, with some describing adaptation ("getting used to it") and others reporting worsening symptoms. Modeling repeated interactions (Figure 3, (Middle)) showed a significant interaction between time and VR experience on log-transformed VRSQ ($\beta = -0.135$, $SE = 0.056$, $t(58) = -2.43$, $p = .018$), alongside a significant main effect of time ($\beta = 0.244$, $SE = 0.089$, $t(58) = 2.75$, $p = .008$). The fitted trajectories indicate that symptoms among lower-experience participants tended to increase over time, whereas those among higher-experience participants showed weaker or negative trends. Notably, even "higher" experience in this sample corresponded to relatively limited VR familiarity. This is particularly relevant for applications such as training simulations, design tools, and games, where users repeatedly resize and may benefit from gradual onboarding or adaptive transition settings.

Transition duration preference. Participants adjusted the transition duration at three distinct points during the experiment (total $N=45$ decisions). As illustrated in Figure 3 (Right), the results indicate a strong *Efficiency Bias* – the sub-second option (0.75s) was the most frequently chosen duration (33% of all decisions). The *Patience*

Threshold (the median preference) was 3.4s. Participants rarely selected transitions longer than seven seconds, suggesting that the "cinematic" appeal of the resizing effect diminishes rapidly against the user's desire to complete the task.

7 Discussion and Implications

Interpreting the null effects. Within the tested parameter range and tasks, we did not detect effects of transition duration or preparedness cueing on VRSQ. These results should be interpreted as bounded evidence, not as proof that transition duration and cueing are irrelevant. Two design decisions likely shape the outcome: (i) movement was locked during resizing to avoid intense sickness induced by concurrent locomotion, and (ii) levels were intentionally short to keep sessions feasible, potentially limiting symptom development and sensitivity to minor effects. Additionally, the preparedness manipulation involved a brief audiovisual warning; alternative cue designs (e.g., stronger multimodal cues or user-confirmed transitions) may yield different outcomes. Furthermore, transition duration was experimentally evaluated at only two options (10s or 3s), and different effects may emerge at intermediate or near-instantaneous ones.

Experience-dependent adaptation suggests onboarding and personalization. The experience-dependent change over repeated exposures suggests that users may not reach stable comfort within a single exposure, and novice users may require different onboarding, pacing, or progressive disclosure of resizing mechanics. Designers should consider staged introductions (e.g., fewer resizing events initially, longer recovery periods) and adaptive transition settings.

The teleportation effect and the "Valley of Boredom." Qualitative feedback challenges the assumption that slower movement always ensures comfort. Seven participants chose the shortest duration (0.75s) specifically to *minimize* cybersickness. They described the sub-second transition as "almost instantaneous" – comparable to a teleportation or cut-scene – which likely bypasses the cybersickness triggers. Participants actively avoided the 2–5s range, which we term the *Valley of Boredom*. They perceived these transition durations as the "worst of both worlds": too fast to be comfortable (inducing sickness), yet too slow to be efficient (inducing boredom). Several participants remarked that they would have chosen slower transitions earlier in the study to enjoy the effect, suggesting that "cinematic scaling" is appropriate, e.g., for tutorials or narrative moments.

Impatience and the "safe harbor." Efficiency was a dominant driver for the majority. Participants described resizing as "idle time," leading to impatience once perceptual novelty faded. However, three participants used the longest duration as a "safe harbor" to mitigate sickness. Notably, even these users indicated their *second* choice would have been the shortest duration, effectively skipping the middle spectrum entirely. This reinforces the finding that users gravitate toward the extremes.

Design implications. Based on the findings, we propose the following design guidelines:

- **Avoid the middle:** Locomotion systems should avoid default transition times in the 2–5s range. This duration appears

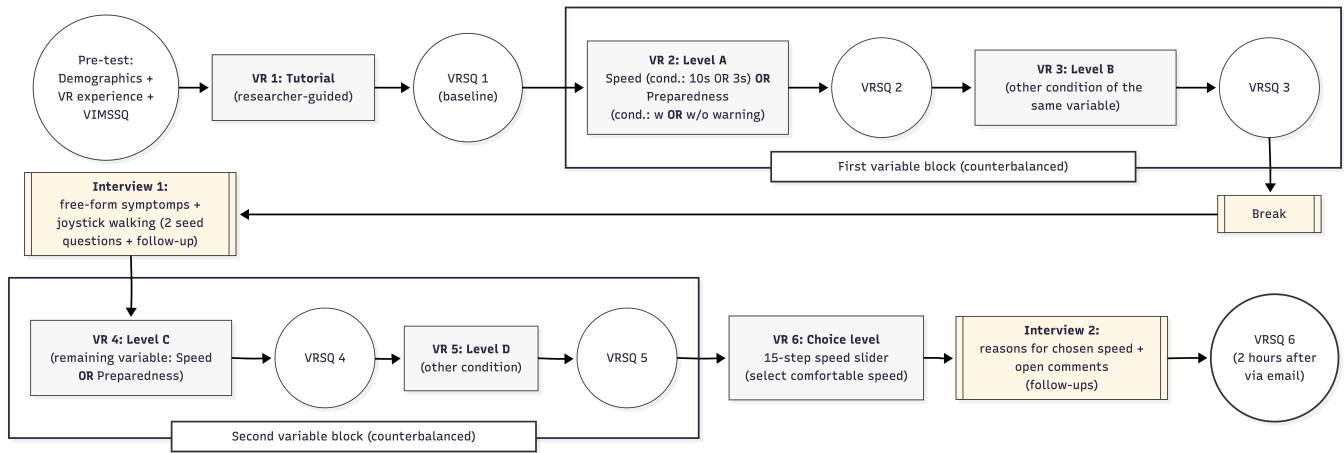


Figure 2: Study procedure timeline: pre-test (demographics/VR experience and VIMSSQ), six VR interactions (tutorial + baseline VRSQ; two levels per factor with counterbalanced order; final transition duration option). VRSQ after each interaction, two interviews (after the break and at the end), and a delayed follow-up VRSQ two hours later via email.

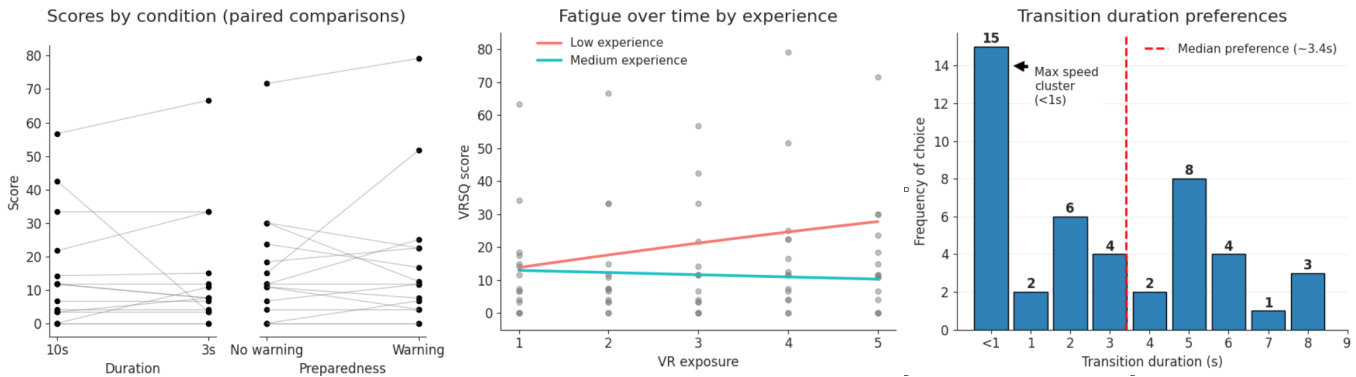


Figure 3: Left: Paired VRSQ scores for each participant across the two conditions (10s and 3s transition durations) and the two preparedness conditions (with/without warning), showing substantial individual variability but no consistent within-participant shift. Middle: VRSQ scores across successive VR exposures (VR Levels), illustrating experience-dependent adaptation: participants with greater prior VR experience tend to show weaker increases (or decreases) in sickness over time compared to less experienced participants. Right: Aggregated distribution of preferred transition durations. The data shows a consistent demand for faster transitions, with the most common choice being the fastest available option (<1s).

to be the “danger zone” where motion is too fast to process comfortably but too slow to perceive as a cut.

- **Snap vs. Glide:** Developers should offer two distinct modes: a “Snap” mode (sub-second scaling) for comfort/efficiency and a “Glide” mode (slow scaling) for immersion/accessibility.
- **Context sensitivity:** Since users cited “boredom” as a primary driver of transition duration, slow transitions should be enforced only when the user has an active task to perform during the scale change (e.g., inspecting an object while shrinking).

Our findings have direct implications for several VR application domains:

- **Precision engineering & CAD applications**, where users must switch between a “god-view” for layout and a 1:1 scale

for detail inspection, designers should default to instant or near-instant (0.75s) transitions to maintain workflow and user satisfaction.

- **VR adventure games** often use slow, cinematic transitions for “epic” scale changes. Our results suggest that for repetitive gameplay mechanics (like our traversal task), slow transitions make users impatient. Therefore, they should be reserved for one-off events rather than frequent mechanics.
- **Accessibility considerations** indicate that faster transitions may serve as a suitable default, while slower alternatives should remain available for users with greater cyber-sickness sensitivity.

7.1 Limitations

The sample size ($N=15$) limits statistical power and may not detect subtle effects of transition duration or preparedness on cybersickness. We experimentally evaluated only two transition durations (10s or 3s), which do not capture the full range of possible transition dynamics. Participants' exposure per level was short to maintain session feasibility; longer or repeated exposure may reveal effects not observable here. Additionally, movement was locked during resizing to isolate transition effects, potentially reducing ecological validity compared to real-world applications. Participants reported that the VRSQ did not fully capture some experienced symptoms, indicating measurement limitations. Additionally, one preparedness level took longer than expected, potentially confounding preparedness with exposure time.

8 Conclusion and Future Work

Grow-shrink interactions offer a compelling VR technique for switching between overview and detailed manipulation in games, visualization, and training environments, but they raise questions about cybersickness and comfort. Our results provide a nuanced perspective on continuous scaling as a locomotion technique and highlight transition duration as an important design parameter influencing user experience. While previous literature [8] focused on the mechanics of scaling, our study isolates the user experience of the transition itself.

We did not detect effects of transition duration (10s or 3s) or preparedness cueing (with or without warning) on VRSQ within the tested setup. In contrast, repeated exposure showed experience-dependent trends, suggesting that adaptation and onboarding may be central to comfort for resizing mechanics. Preference results indicate a non-linear comfort landscape, with users selecting extremes. The convergence of quantitative and qualitative data suggests a “bimodal” preference driven by a conflict between efficiency and sensory comfort. We found that users are “impatient giants”: they enjoy the capability to resize, but they demand that it happens at the speed of thought.

Future work should investigate if the *Valley of Boredom* shifts in dynamic contexts where users can move or dodge while resizing, potentially making intermediate transition duration more engaging. The study should be replicated with larger samples, longer and more controlled exposures, richer transition duration sampling across the spectrum, and complementary objective measures (e.g., physiological indicators). Particularly promising is also the exploration of alternative cueing strategies and transition designs that preserve agency while limiting discomfort.

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